RUNGTA INTERNATIONAL SKILLS UNIVERSITY

SCHOOL OF INFORMATION TECHNOLOGY
BACHELOR OF COMPUTER APPLICATION

Student Name:	SHIVAM KUMAR VERMA
PS No. / Project ID:	26
Project Title:	Population Data Visualizer
Platform / Language:	Python 3 (Pandas, Matplotlib)
GitHub Repository:	https://github.com/shivamverma2019-ctrl/Population-Data-Visualizer

Project Problem Statement

Analyse city population data from a CSV file to find the total population, the largest city, and each city's growth between years. Clean invalid or duplicate rows and dynamically visualize the output using professional charts.

Implemented Test Cases Verification

Test Case	Input Condition	Expected & Verified Output
TC 1: Normal Input	Multi-year records for Raipur & Durg (2020-2024)	Total population (2024): 1700000 Largest city: Raipur (1100000) Fastest growing: Durg (+20.00%)
TC 2: Boundary Input	Single row / Single year data baseline	Total population: 1100000 | Largest city: Raipur Growth rate statement: *'Growth: not available (only one year of data)'* handled smoothly.
TC 3: Invalid Input	Corrupted values ('abc') & Negative populations (-500)	Row 2 skipped: invalid Population 'abc'. Row 3 skipped: Population must be positive. Valid rows parsed summary display.
TC 4: Multi-Variable	Console Menu System Selection: Option 2	Calculates proportional growth splits and rendering active twin Matplotlib figures (Bar Chart + Trend Line Graph).

Bare Minimum Requirements Met

1. Write the solution in Python 3 (one .py file) and run it from the console.
2. Load the CSV file using Pandas and clean invalid or duplicate rows.
3. Use groupby() to find the latest population and the growth percentage of every city.
4. Show the total population, the largest city, and the fastest growing city.
5. Plot at least two Matplotlib charts (line and bar) with title and labels.